Defeating the Prefix-Cache Penalty

KV-Aware Routing & Disaggregated Serving for Agent Fleets on GKE

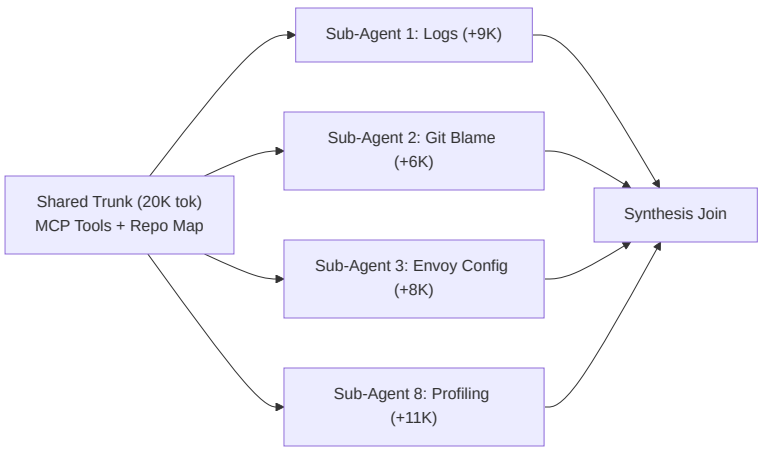
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Gateway API Inference Extension · `llm-d` · `vllm-torchtpu` · GKE Cloud TPU

Chatbots Are Lines. Agent Fleets Are Trees.

Chatbot Workload (2024)

- Single linear user session
- Human typing pace (15–30s think time)
- Small system prompt (~1K tokens)
- Predictable HBM footprint



SWE / SRE Agent Fleet (2026)

- **20K Shared Trunk:** Repo AST + 120 MCP schemas + incident context
- **Fork-Storm (`t = 0ms`):** Orchestrator spawns **8 parallel sub-agents** simultaneously
- **Tool Bursts:** Sub-agents ingest 10K `kubectl logs` / `git diff` dumps
- **Join Turn:** Orchestrator synthesizes branches

Failure Mode 1: Stateless K8s & The Scatter Penalty

Standard `kind: Service` scatters forked sub-agents randomly across N replicas:

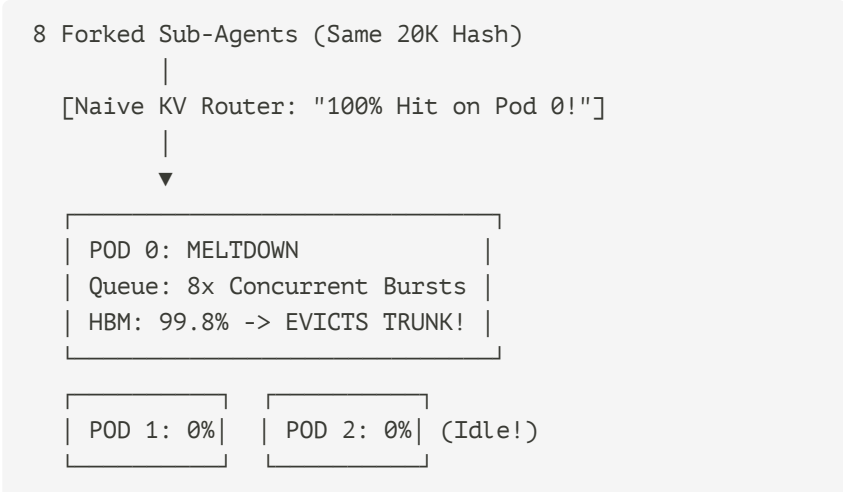
- **1/N Cache Collapse:** Pod A, Pod B, Pod C, and Pod D *all* recompute the identical 20K parent trunk on their Matrix Multiply Units (MXUs).
- **HBM Duplication:** Identical 20K FP8 KV blocks occupy $N\times$ High-Bandwidth Memory across the cluster.
- **Prefill-Decode Interference:** When Sub-Agent 3 dumps 10K log lines into an aggregated pod, compute-bound prefill stalls memory-bound decode for every co-located agent (P99 ITL spikes past 115 ms).

```
Orchestrator Fork (8x 20K Trunk) ──► K8s ClusterIP Service (Round-Robin)
   ├──► Pod 0: Recomputes 20K Trunk (Cache Miss) + Stalls Active Decodes
   ├──► Pod 1: Recomputes 20K Trunk (Cache Miss) + Stalls Active Decodes
   └──► Pod 2: Recomputes 20K Trunk (Cache Miss) + Stalls Active Decodes
```

Failure Mode 2 (Plot Twist): The Prefix-Herd Trap

What happens when you enable **Naive Longest-Prefix Affinity Routing**?

1. **100% Match = Single-Pod Stampede** All 8 forked sub-agents share the *exact* 20K parent block hashes on **Pod 0**. Naive KV routing sends **all 8 concurrent branches to Pod 0**.
2. **Queue Explosion** Pod 0's prefill queue saturates while Pods 1…3 sit idle—spiking **P95 TTFT higher than Round-Robin**.
3. **Self-Inflicted HBM Thrash** As the 8 diverging branches append 8K–11K tool outputs on Pod 0, they overflow HBM and **LRU-evict their own 20K parent trunk** before the Join turn!



Why Routing & Disaggregation Are Inseparable

To defeat *both* Scatter and Herd penalties on agent trees, control plane and data plane must cooperate:

Problem Layer	Naive Aggregated Cluster	**FleetKV + TorchTPU Disaggregated Stack**
Shared 20K Trunk	Recomputed N times *or* herds onto 1 pod	**Computed ONCE** on Prefill TPU; pinned in Tier-2 Host DRAM
8x Branch Fan-Out	Queues serially or evicts HBM	**Admit-if-Fits** spreads branches across Decode TPUs
10K Tool-Log Burst	Stalls active decode ITL	Isolated on Prefill MXUs; Decode ITL stays flat
KV State Mobility	Trapped inside single pod HBM	Async **TPU Sync (** `TPUConnectorHMA` **)** streams over gVNIC

Control Plane: Fit-Aware Tree Routing on GAIE

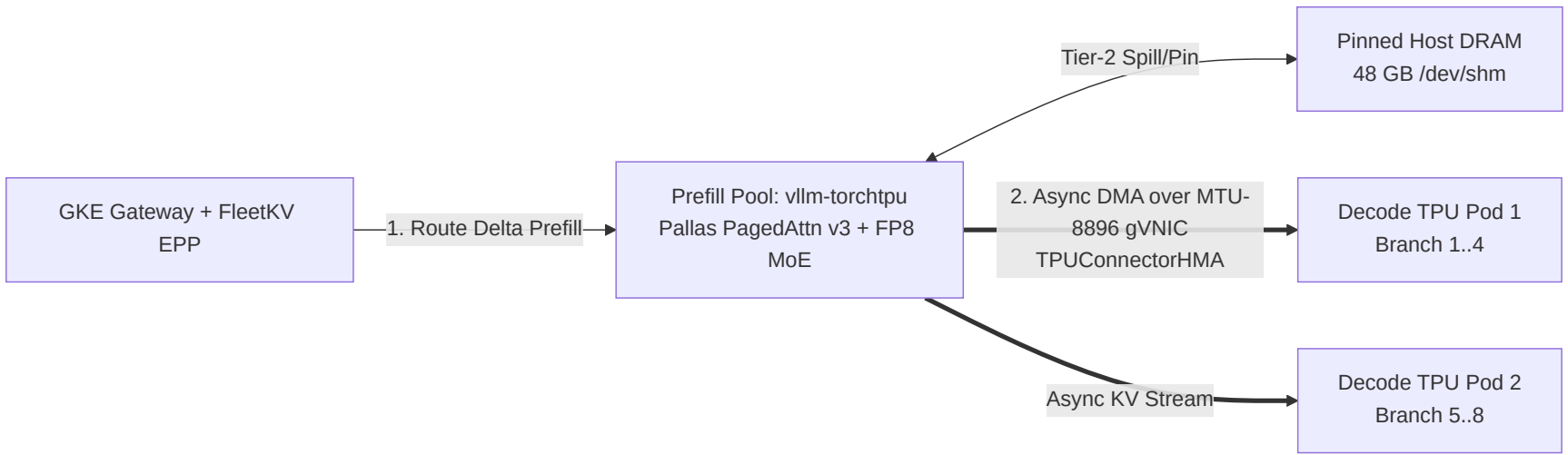
Built natively on **Gateway API Inference Extension (`v1`)** + Envoy `ext_proc` + ZeroMQ KV Events:

```go
// FleetKV Composite Admission & Scoring inside GAIE ext_proc
func ScoreEndpoint(pod *PodState, req *AgentBranchReq) float64 {
    // 1. Hard HBM Fit Gate: reject pod if branch horizon risks evicting hot trunk blocks
    if pod.FreeHBMBlocks < req.EstimatedDeltaBlocks + SafetyReserveBlocks {
        return RejectOverflowSpillToPeer
    }
    // 2. Balance exact ZMQ-tracked block residency against live prefill queue wait
    marginalSavedMs := float64(pod.ExactMatchedBlocks(req.BlockHashes)) * BlockPrefillCostMs
    queuePenaltyMs  := float64(pod.InFlightPrefillTokens) / PodThroughputTokPerMs
    return marginalSavedMs - queuePenaltyMs
}
```

- **Zero Guessing:** `vllm-torchtpu` streams `BlockStored` / `BlockRemoved` over ZMQ (`tcp://*:5557`) to keep an exact cluster-wide radix tree of 256-token blocks.
- **Fit-Gated Fan-Out:** Prevents HBM self-eviction by gating admission on free KV block horizon and coordinating P/D handoff headers.

Data Plane: `vllm-torchtpu` + `TPU Sync` on GKE

100% upstream PyTorch models running natively on **GKE Trillium (`ct6e-standard-4t`) / Ironwood TPUs**:



- **Zero Model Rewrites:** `torch.compile(backend="tpu")` lowers PyTorch into Pallas Ragged Paged Attention & fused FP8 MoE kernels.
- `TPUConnectorHMA` **(** `TPU Sync` **):** Asynchronous device-to-device & host-DRAM KV transfer overlaps network DMA with active decode steps—cutting fan-out transfer overhead below 35 ms.

Live 3-Stage Traffic Flip on GKE

8x Parallel Sub-Agent Fork-Storms · `Qwen3-30B-A3B-FP8` · Equal 8-Chip TPU Budget

1. `weight: 100` → **Standard K8s** `Service` **(Round-Robin)** — *Watch Scatter Misses*
2. `weight: 100` → **Naive KV-Sticky** `InferencePool` — *Watch Prefix-Herd Queue & Evictions*
3. `weight: 100` → `FleetKV` **Fit-Aware +** `vllm-torchtpu` **Disagg** — *Watch TTFT & ITL Collapse*

Equal-Hardware Benchmark Scorecard

Same **8 TPU v6e Chips** (`2x ct6e-standard-4t`) · **64 Active Agent Branches** (8 Orchestrators × 8-Way Fork)

Metric	1. K8s Round-Robin	2. Naive KV-Sticky	**3. FleetKV + TorchTPU P/D**
Effective Prefix Hit Rate	19.4 %	52.8 % *(thrashing)*	**91.6 %**
Median TTFT (P50)	2,710 ms	1,180 ms	**315 ms (-88%)**
Fork-Burst Tail TTFT (P95)	3,420 ms	3,890 ms *(herd stall!)*	**485 ms (-86%)**
Decode Jitter (P99 ITL)	116 ms	98 ms	**27 ms (-77%)**
Cluster Output Goodput	1.0x	1.35x	**2.55x**

Three Production Scars (Save Your Weekend)

1. Pin `PYTHONHASHSEED=0`

Python salts process hashes by default. Without `PYTHONHASHSEED="0"` on **every** vLLM pod and router, identical 256-token blocks hash differently and P2P KV hits silently drop to `0%`.

2. TPU `--block-size 256`

GPU recipes default to `64`. On TPU v6e, block size `64` overflows Scalar Memory (SMEM) in Pallas paged attention. Set `--block-size 256` in vLLM **and** `blockSizeTokens: 256` in EPP.

3. ZMQ Role Filter + MTU 8896

Filter EPP ZMQ discovery to `llm-d.ai/role=prefill` so routers don't dial decode pods, and set GKE gVNIC VPC `mtu: 8896` so `TPU Sync` avoids packet fragmentation.

Stop Routing Agent Trees Like Stateless REST

Key Takeaways

- **Agents fork in bursts:** Naive sticky routing trades cache misses for hot-pod stampedes and HBM self-eviction.
- **Route on Fit + Marginal KV Delta:** Gate admission on HBM block headroom via GAIE `ext_proc` + ZeroMQ KV events.
- **Disaggregate on TorchTPU:** Compute trunks once on Prefill, pin in Host DRAM, fan out deltas via `TPU Sync`.

Apache-2.0 Open Source Repo

`github.com/rkaranjai/kubecon26-fleetkv-gke`

Includes:

- Go GAIE `ext_proc` `FleetKV` Router + ZMQ Indexer
- GKE TPU v6e Terraform + `DisaggregatedSet` YAML
- Local Docker Twin (rehearse on laptop!) + TUI